MIL-A- 48100 (AR)
7 July 1977
SUPERSEDING
See Section 6

MILITARY SPECIFICATION

ADAPTION KIT, ATOMIC WARHEAD TO GUIDED MISSILE, TRAINING: XM255

This specification is approved for use by all Departments and agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of training adaption kit entitled Adaption Kit, Atomic Warhead to Guided Missile, Training: XM255 which is used for training on the Sprint warhead.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-S- 48893	- Shipping and Storage Container: XM543E1.
MIL-C- 50952	- Type X-W Trainer Cables.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables For Inspection by Attributes.
MIL-STD-1168	- Lot Numbering of Ammunition.

SUPPLEMENTARY QUALITY ASSURANCE PROVISIONS

PICATINNY ARSENAL

SQAP 9289206	- Arm-Safe Device, Atomic Weapon, Training: XM141
SQAP 9247210	- Adaption Kit, Sub-Assembly Atomic Weapon, Training: XM209

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document, should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAN-Q, Dover, New Jersey 07801, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document, or by letter.

FSC 1135

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:13Z
Check the source to verify that this is the current version before use.

DRAWINGS

ORDNANCE CORPS

8888397 - Socket Head Cap Screw.
9247210 - Adaption Kit, Sub-Assembly Atomic Weapon, Training: XM209.
9257234 - Shipping and Storage Container, XM543E1, Packing and Marking, Package 1 of 3, for Adaption Kit, Atomic Warhead to Guided Missile.
9272819 - Washer Flat, Modified.
9273595 - Test Set Atomic Weapon: T-4197.
9274150 - Adaption Kit, Atomic Warhead to Guided Missile, Training: XM255.
9275835 - Packing and Marking for Box, Fibreboard, Packing, Outer Pack 2 of 3 for Adaption Kit, Atomic Warhead to Guided Missile.
9275836 - Packing and Marking for Box, Fibreboard, Packing, Pack 3 of 3 for Adaption Kit, Atomic Warhead to Guided Missile.
EL-9272819 - Equipment List.
EL-9257234 - Equipment List.
EL-8888397 - Equipment List.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The training adaption kit shall comply with all requirements specified on the drawing (dwg) 9274150, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. The contractor shall assign serial numbers as required by drawings. The contractor may select any number which complies with the drawing requirement. These numbers shall not be duplicated.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:13Z
Check the source to verify that this is the current version before use.

3.5 Continuity loop.

3.5.1 Continuity loop-Adaption Kit, Sub-Assembly, Atomic Weapon, Training: XM209 (see dwg 9247210). The continuity loop in the Adaption Kit, Subassembly, XM209 shall show a continuous circuit when tested as specified in 4.4.1.

3.5.2 Continuity loop-Arm-Safe Device: XM141 (see dwg 9289206). The continuity loop of the Arm-Safe Device shall show a continuous circuit when tested as specified in 4.4.1.

3.6 Major components. For major components to be used in the Adaption Kit, Atomic Warhead to Guided Missile, Training: XM255 objective evidence (see 6.2) indicating that the following components conform to their representative specifications and or drawings shall be made available to the Government representative.

a. MIL-S-48893 (PA) - Shipping and Storage Container
b. MIL-C-50952 (PA) - Type X-W Trainer Cables.
c. SQAP-9289206 - Arm-Safe Device Atomic Weapon Training: XM141.
d. SQAP-9247210 - Adaption Kit, Subassembly Atomic Weapon, Training: XM209.

3.7 Workmanship.

3.7.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.7.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.7.3 Burrs. Parts shall be free from burrs.

3.7.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.7.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.7.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:13Z
Check the source to verify that this is the current version before use.

perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 1 complete assembly (see dwg 9274150) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table I. Tests shall be conducted in order specified in Table I.

Table I

First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of Characteristics/ Defects (see 4.3.2.1)		X
2. Continuity loop (see 3.5)		X

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:13Z
Check the source to verify that this is the current version before use.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor defects/characteristics table (see Table II and Table III), and inspected for all defects/characteristics in the applicable classification. Any defects/characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of defects/characteristics in each classification need not be followed as long as all defects/characteristics inspections are performed.

Table II
Major Defects/Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

Table III
Minor Defects/Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

4.3.2.1 Classification of defects/characteristics (see 3.2). Critical, major and minor defects/characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Socket head cap screw (see dwg 8888397).

Categories	Characteristics/Defects	Method of Inspection
Critical:	None defined.	
Major:		
	101. Objective evidence (see 6.2) that item complies with drawing	Visual
	102. Length of thread	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:13Z
Check the source to verify that this is the current version before use.

Major: (Cont).

103. Thread damaged.....Visual
104. Evidence of poor workmanship (see 3.7.2).....Visual

Minor: None defined.

4.3.2.1.2 Washer flat, special purpose (see dwg 9272819).

Categories	Characteristics/Defects	Method of Inspection

Critical: None defined.

Major:
101. Objective evidence (see 6.2) that part conforms to AN 960-C10.....Visual
102. Outside diameter.....Gage
103. Evidence of poor workmanship (see 3.7.2).....Visual

Minor: None defined.

4.3.2.1.3 Bolt, twelve, external wrenching (MS-21250-04018): Washer, countersunk (MS20002-04; washer, flat (MS9321-09); screw, self locking, pan head, full thread (NAS 1190-3P10L); screw, cap socket head (MS 16998-26): and washer, flat, for wood (AN 970-3), prior to packing.

Categories	Characteristics/Defects	Method of Inspection

Critical: None defined.

Major:
101. Objective evidence (see 6.2) that components comply with applicable government standards.....Visual

Minor: None defined.

4.3.2.1.4 Adaption kit, sub-assembly atomic weapon, training: XM209, (see dwg 9247210).

Categories	Characteristics/Defects	Method of Inspection

Critical: None defined.

Major:
101. Objective evidence (see 6.2) that components complies with 3.6.....Visual
102. Serialization missing or improper (see 3.4).....Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:13Z
Check the source to verify that this is the current version before use.

Minor:

201. Evidence of poor workmanship (see 3.7).....Visual

4.3.2.1.5 Packing of socket head cap screw (8888397); washer flat, modified (9272819); bolt, twelve point, external wrenching (MS-21280-04018); washer countersunk (MS 20002-04); washer, flat (MS 9321-09); screw locking, pan head, full thread (NAS 1190-3P10L); screw, cap, socket head (MS 16998-26); and washer, flat, for wood (AN 970-3).

Categories	Characteristics/Defects	Method of Inspection
Critical:	None defined.	
Major:		
	101. Improper amount of hardware in envelopes (see sheet 1 of dwgs 9257234, 9275835 or 9275836 as applicable	Visual
	102. Improper number (4) of envelopes in barrier bag (prior to sealing barrier bag) (see dwg 9257234)	Visual
	103. Barrier bag improperly sealed (see dwg 9257234)	Visual
	104. Marking on envelopes missing, illegible, or does not correspond to hardware in envelope	Visual
	105. Marking on barrier bag missing, improper or illegible	Visual
Minor:	None defined.	

4.3.2.1.6 Packing of package 1 of 3 (see dwg 9257234).

Categories	Characteristics/Defects	Method of Inspection
Critical:	None defined.	
Major:		
	101. Talc missing from rubber surfaces of frame assembly and retaining plate	Visual
	102. Adaption kit sub-assembly improperly seated in frame assembly	Visual
	103. Bags (4) of molecular sieve desiccant not in basket	Visual
	104. Barrier bag with mounting hardware not secured to retaining plate with tape	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:13Z
Check the source to verify that this is the current version before use.

Major: (Cont).

105.	Nuts holding retaining plate not torqued sufficiently	Gage
106.	Ground cable not secured to adaption kit sub-assembly	Visual
107.	Cables improperly placed and secured to cable bracket	Visual
108.	Swing latches secured improperly	Visual/Manual
109.	Metallic seal missing or improperly crimped	Visual
110.	Marking missing, illegible or improper	Visual
111.	Evidence of poor workmanship (see 3.7.2)	Visual
112.	Packing or cushioning material missing	Visual

Minor: None defined.

4.3.2.1.7 Packages 2 of 3 and 3 of 3 (see dwg 9275835 and 9275836 respectively).

Categories	Characteristics/Defects	Method of Inspection

Critical: None defined.

Major:

101.	Objective evidence (see 6.2) that the items packed in packages 2 and 3 of 3 comply with 3.6	Visual
102.	Packages improperly sealed	Visual
103.	Marking improper, missing or illegible	Visual
104.	Serial numbers on all packages do not correspond	Visual

Minor: None defined.

4.3.3 Testing. The following test shall be performed:

Test	Test Classification
a. Continuity loop	Major

This test is not destructive and samples so tested may be returned to the lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:13Z
Check the source to verify that this is the current version before use.

4.3.3.1 Continuity loop. Each training adaption kit shall be tested in accordance with 4.4.1. If any item fails to comply with 3.5, the item shall be rejected and removed from the lot.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major defects/characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see ELS- 8888397; 9957234, and 9272819). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior touse on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor defects/characteristics and tests. Acceptance inspection equipment for minor defects/characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Continuity loop. The continuity loops of the adaption kit and the Arm-Safe Device shall be tested using the Test Set, Atomic Weapon T-4197 and the following test procedure.

Note: This document constitutes complete instructions for use of the T-4197 Atomic Weapon Test Set.

4.4.1.1 General Instructions:

a. Perform Pre-Test Unit Verification
b. Perform Pre-Test Cable Verification (as appropriate).
c. Perform Test on AK (as appropriate).
d. Perform Post-Test Cable Verification (as appropriate).
e. Perform Post-Test Unit Verification.

Note: Any indication other than those called for in a specific step indicates a malfunction of the test equipment, or of the system being tested. Further tests utilizing equipment must not be made until cause of malfunction is determined and corrected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:13Z
Check the source to verify that this is the current version before use.

4.4.1.2 Sequence of Operations:

a. Pre-Test Unit Verification.
b. Pre-Test Cable Verification Sprint XT4197-1 Cable.
c. Sprint AK Status Circuit Test.
d. Post-Test Cable Verification Sprint XT4197-1 Cable.
e. Pre-Test Cable Verification Sprint XT4197-3 Cable.
f. Arm Safe Device Circuit Test.
g. Post-Test Cable Verification Sprint XT4197-3 Cable.
h. Post-Test Unit Verification.

4.4.1.3 Pre-Test Unit Verification:

a. Open Cover of Test Set.
b. Turn S1 on Test Set to 1-OFF.
c. Depress DS1, while rotating generator knob. DS1 should light.
d. Turn S1 to 3 - GEN AND LAMP TEST.
e. Turn generator knob. DS1 should light.
f. Turn S1 to 4 - SELF TEST.
g. Turn generator knob. DS1 should light.
h. Turn S1 to 5 - DISC TEST.
i. Turn generator knob. DS1 should not light.
j. Depress DS1, while rotating generator knob. DS1 should light.
k. Turn S1 on Test Set to 1-OFF.

4.4.1.4 Pre-Test Cable Verification - Sprint XT4197-1 Cable:

a. Open lid of test set cover containing XT4197-1 Sprint Cable.
b. Remove cable and close lid.
c. Remove dust covers from J1-COMMON and J3-SPRINT on panel of test set.
d. Remove dust covers from P1 and P3 ends of XT4197-1 Cable.
e. Attach P1 end of cable to J1 on panel.
f. Attach P3 end of cable to J3 on panel.
g. Turn S1 to 7 - SPRINT B.
h. Turn generator knob. DS1 should light.
i. Turn S1 to 8 - SPRINT C.
j. Turn generator knob. DS1 should light.
k. Turn S1 to OFF.
l. Detach P3 end of cable from panel.

4.4.1.5 Sprint AK Status Circuit Test:

a. Remove environmental protective cap from adaption kit receptacle J2.
b. Connect test cable XT4197-1, P3, to adaption kit receptacle J2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:13Z
Check the source to verify that this is the current version before use.

c. Turn test set S1 to 7 - SPRINT B.
d. Turn generator knob. DS1 should light.
e. Turn test set S1 to 8 - SPRINT C.
f. Turn generator knob. DS1 should light.

4.4.1.6 Post-Test Cable Verification - Sprint XT4197-1 Cable:

a. Attach P3 of XT4197-1 Sprint cable to J3 on panel.
b. Turn S1 to 7 - SPRINT B.
c. Turn generator knob. DS1 should light.
d. Turn S1 to 8 - SPRINT C.
e. Turn generator knob. DS1 should light.
f. Turn S1 to 1-OFF.
g. Detach P3 end of cable from J3 on panel.
h. Detach P1 end of cable from J1.
i. Place dust covers on P1 and P3 ends of cable.
j. Place dust covers on J1 and J3 on panel.
k. Open lid of test set cover which is to contain XT4197-1 Sprint Cable.
l. Place cable inside and close lid.

4.4.1.7 Pre-Test Cable Verification - SPRINT XT4197-3 Cable:

a. Open lid of test set cover containing XT4197-3 Sprint Cable.
b. Remove cable and close lid.
c. Remove dust covers from J1-COMMON and J4-SPRINT on panel of test set.
d. Remove dust covers from P1 and P4 ends of cable.
e. Attach P1 end of cable to J1 on panel.
f. Attach P4 end of cable to J4 on panel.
g. Turn S1 to 9 - SPRINT D.
h. Turn generator knob. DS1 should light.
i. Turn S1 to 1-OFF.

4.4.1.8 Arm Safe Device Circuit Test:

a. Remove environmental protective cap from Arm-Safe Device J3.
b. Connect test cable XT4197-3, P4 end to Arm-Safe Device J3.
c. Turn test set S1 to 9 - SPRINT D.
d. Turn generator knob. DS1 should light.
e. Turn test set S1 to 1-OFF.
f. Disconnect test cable XT4197-3, P4 end from Arm-Safe Device J3.
g. Reinstall environmental protective cap on Arm-Safe Device J3.

4.4.1.9 Post-Test Cable Verification - SPRINT XT4197-3 Cable:

a. Attach P4 end of XT4197-3 Sprint Cable to J4 on panel.
b. Turn S1 to 9 - SPRINT D.
c. Turn generator knob. DS1 should light.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:13Z
Check the source to verify that this is the current version before use.

d. Turn S1 to 1-OFF.
e. Detach P4 end of cable from J4 on panel.
f. Detach P1 end of cable from J1 on panel.
g. Place dust covers on P1 and P4 ends of cable.
h. Place dust covers on J1 and J4 on panel.
i. Open lid of test set cover which is to contain XT4197-3 Sprint Cable.
j. Place cable inside and close lid.

4.4.1.10 Post-Test Unit Verification:

a. Turn S1 to 1-OFF.
b. Depress DS1 while rotating generator knob. DS1 should light.
c. Turn S1 to 3 - GEN AND LAMP TEST.
d. Turn generator knob. DS1 should light.
e. Turn S1 to 4 - SELF TEST.
f. Turn generator knob. DS1 should light.
g. Turn S1 to 5 - DISC TEST.
h. Turn generator knob. DS1 should not light.
i. Depress DS1 while rotating generator knob. DS1 should light.
j. Turn S1 on test set to 1-OFF.
k. Close cover of test set.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The training adaption kit shall be packaged in accordance with dwgs 9257234, 9275835 and 9275836 using the specified moisture barrier.

5.1.2 Level C. The training adaption kit shall be packaged in accordance with dwgs 9257234, 9275835 and 9275836 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The training adaption kit shall be packed in accordance with dwgs 9257234, 9275835 and 9275836.

5.3 Marking. Marking shall be as specified in accordance with dwgs 9257234, 9275835, and 9275836.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAN-I, Dover, New Jersey 07801.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:13Z
Check the source to verify that this is the current version before use.

6.4 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-2225A dated 6 November 1974.

Custodian:
Army -(AR)

Preparing activity:
Army - (AR)

Project No. 1135-A227

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:13Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD – 314

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
US Army Armament R&D Command
ATTN: DRDAR-QAN-Q
Dover, NJ 07801

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:13Z
Check the source to verify that this is the current version before use.